# Documentation

## Package : Modular Medieval Citizens

version. 1.1

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# Short Description

Medieval peasants with 13 modular body parts, 23 changeable color masks and customizable face blend-shapes with lipsync. Endless possibilities creating new characters with random face generation. Clothes and hair using simulation.

# Long Description

Preview video : coming soon

Medieval peasants with 13 modular body parts, 23 changeable color masks and customizable face blend-shapes with lipsync. Endless possibilities creating new characters with random face generation. Clothes and hair using simulation. Easy to create new character using Character Creator Blueprint via UE variables. Camera in TPP and FPP mode.

Modular Medieval Citizen pack contain:

- **Character Creator Blueprint** with endless possibilities - 111 variables you can change!
- **Changeable 13 body parts** (helmet, hood, hair, beard, chest, shoulder, gloves, skirt, pants, boots, tools, apron, jacket) with different models and materials
- **23 body parts** where you can change HSL color (Hue, Saturation, Lightness)
- **Face Creator** using Character BP variables (blendshapes and material instance)
- age and racial control on face
- simple **auto lip sync system** (10 lip blendshapes [AI, E, CDGKNRSTYZ, FV, O, U, L, MBP, WQ, rest] with simple script to auto lip sync on String input)
- naked models divided into parts (hands, arms, chest, all) - auto change visibility according to armor visibility
- body proportion control
- Hair and Cloth simulation
- rigged and scaled to epic skeleton - easy to retarget any animation using Epic skeleton
- 2 camera types - TPP and FPP
- 4 sockets for (shields) and tools
- rig contain twist bones - a lot better transformations in animations
- Character blueprint - **ready to play**
- easy way to add custom meshes and materials to randomization BP
- animations (Epic animations, modified inside Persona)
- documentation file with tutorial!

Technical Details:
Scaled to Epic skeleton: Yes
Rigged: Yes
Rigged to Epic skeleton: Yes
Animated: Yes
Number of characters: 1
Vertex counts of characters: lod0 = ~25k (fully equipped, full naked body + 10k)
Vertex counts of characters: lod1 = ~7k (fully equipped, full naked body + 2.5)
Texture Resolutions: 1024x1024 - 4096x4096
**183 PBR** textures (albedo, normal, RGB)
**61 models** with 2 levels of details
Optimized textures using one RGB texture with channels for - roughness, AO, metallic.
Textures in DXT1 format - 2x less VRAM usage (normal map BC5
Number of Animations: 6
Animation types (Root Motion/In-place): 2/4

Engine Compatibility: 4.12 - 4.19
Intended Platform: Windows, Mac, PS4, iOS, Android, Xbox One
Platforms Tested: PC
Important\Additional Notes:
Price can change in the next updates when I will add additional textures and models.
**You can mix this package with Modular Medieval Warrior.**

# Overview

## Name convention:

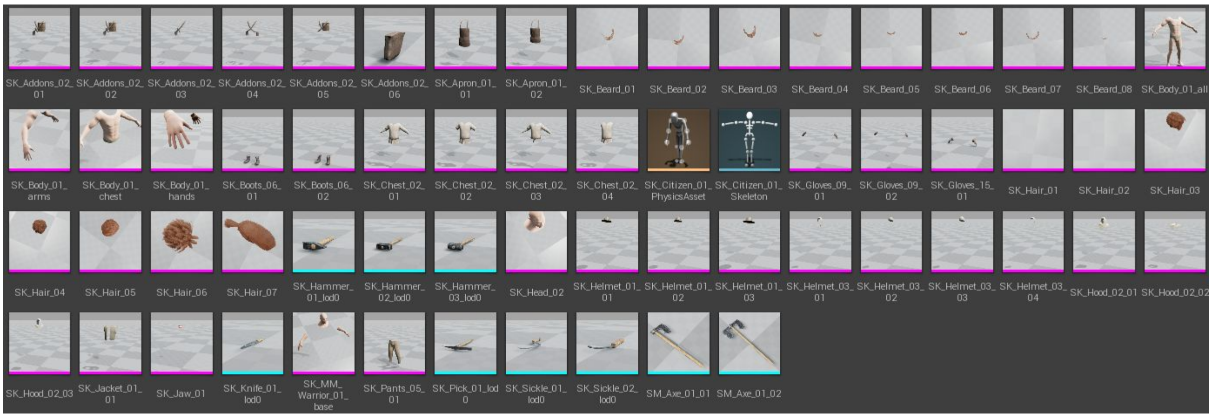
Models:
SK_objName_type_version
Example: SK_Chest_01_03
SK - skeletal mesh, SM - static mesh
Chest - object name
01 - type
03 - version

Textures:
T_objName_type_texType_version
TexType are: diff - diffuse/albedo, norm - normal map, RGB (roughness, AO, metallic), masks - RGB mask used for coloring
Example: T_Chest_01_01_diff

## Folder structure:

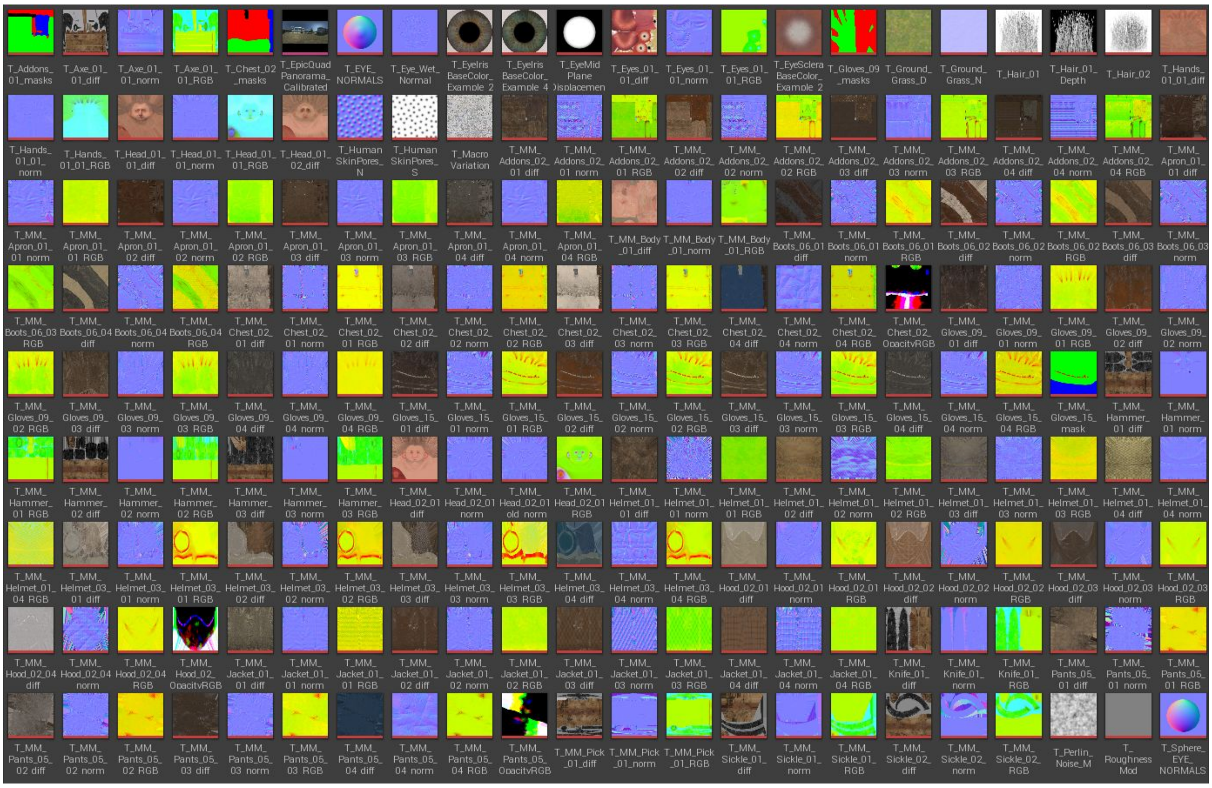
Content
  MedievalWarrior_01
  - Animations
  - Blueprints
  - Materials
  - Meshes
  - Textures

# Models



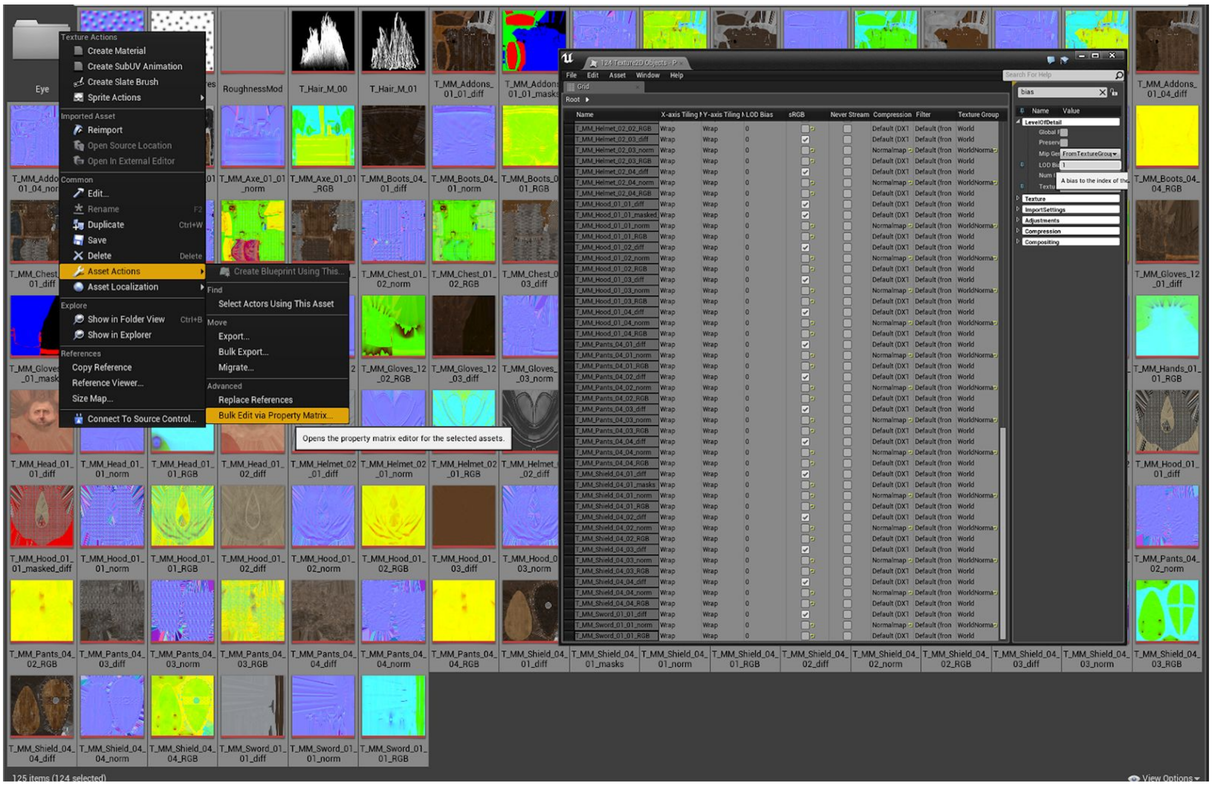

60 models with lod0 and lod1:
4 body: all, arms, chest, hands
6 Addons (Shoulders)
2 Aprons
9 Tools - 3 hammers, 1 knife. 1 pick, 2 sickles, 2 axes
8 Beards
2 Boots
4 Chests - 2 materials, skirt and chest, skirt using apex cloth
1 Jacket
3 Gloves - 2 types
7 Hairs - 03-07 using apex cloth for simulation
7 Helmets - 2 types
3 Hoods
1 Pants

# Textures



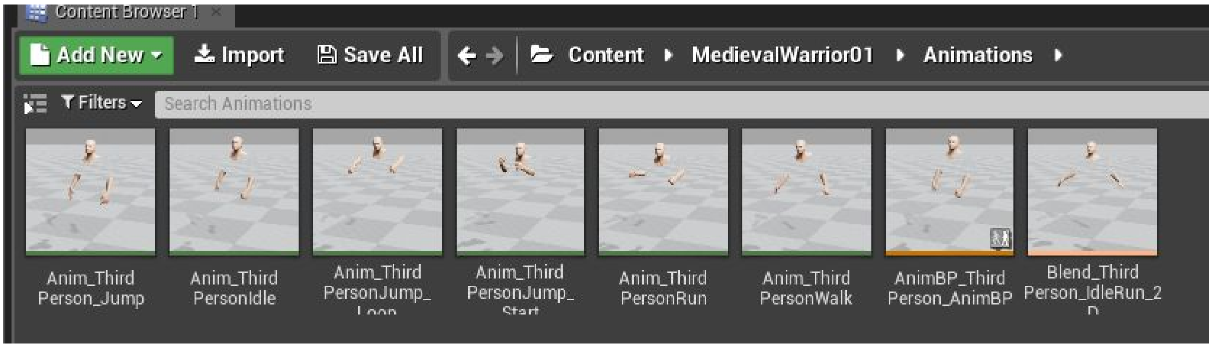
Eye - Epic Eye Textures
2x - HumanSkinPores - Epic Skin additionally textures
3x -Axe - 1x3  (diff, norm, RGB) - 4096x1024
13x -  Addons(Shoulders) - 4x3 (diffuse, normal, RGB) +1 mask(RGB) - 2048x1024
12x - Aprons - 4x3 (diffuse, normal, RGB) - 2048x2048
3x - Body - 1x3(diffuse, normal, RGB)  - 4096x4096
12x - Boots - 4x3 (diffuse, normal, RGB) - 2048x2048
14x Chest - 4x3 (diffuse, normal, RGB) + 1 Opacity (RGB) + 1 mask(RGB)-  4096x4096
25x Gloves - 8x3 (diffuse, normal, RGB) +1 mask(RGB) - 12x 2048x1024, 12x 2048x2048
9x Hammers -  3x3 (diffuse, normal, RGB) - 6x  2048x1024, 3x 2048x512
24x Helmets -  8x3 (diffuse, normal, RGB) - 12x 2048x1024, 12x 2048x2048
3x - Eyes_01_01_diff - 1x3 (diffuse, normal, RGB) - 2048x2048 - it’s used for Jaw
2x - Hair - 1x2  (opacity, depth) - 1024x1024
3x - Hands - 1x3  (diff, norm, RGB) - 2048x2048
7x - Head - 4+3  (diff1, diff2, norm, RGB) - 4096x4096
13x - Hood- 4x3 (diffuse, normal, RGB)  +1 mask(RGB)  - 2048x1024
12x - Jacket - 4x3 (diffuse, normal, RGB) - 4096x2048
3x Knife- 1x3  (diff, norm, RGB) - 1024x1024
13x - Pants- 4x3( (diff, norm, RGB) + 1 Opacity (RGB)  - 2048x2048
3x Pick - 1x3  (diff, norm, RGB) - 2048x2048
6x Sickle1x3  (diff, norm, RGB) - 2048x2048

You can easily downscale texture with “LOD Bias” texture variables set to 1 or higher:




*Select all texture -> Right Mouse -> Asset Actions -> Bulk Edit via Property Matrix -> Change “LOD Bias” to 1 or higher*

# Animations




Jump, Jump_Loop, Jump_Start, Idle, Run, Walk - modified within Persona - epic animations
MM_ThirdPerson_IdleRun_2D - blendspace for walk - run blending

*Skeleton tree setup for retargeting.*

Retarget Manager configured for retargeting using humanoid rig.
Character is in Epic Tpose as a Ref Pose for best retargeting.

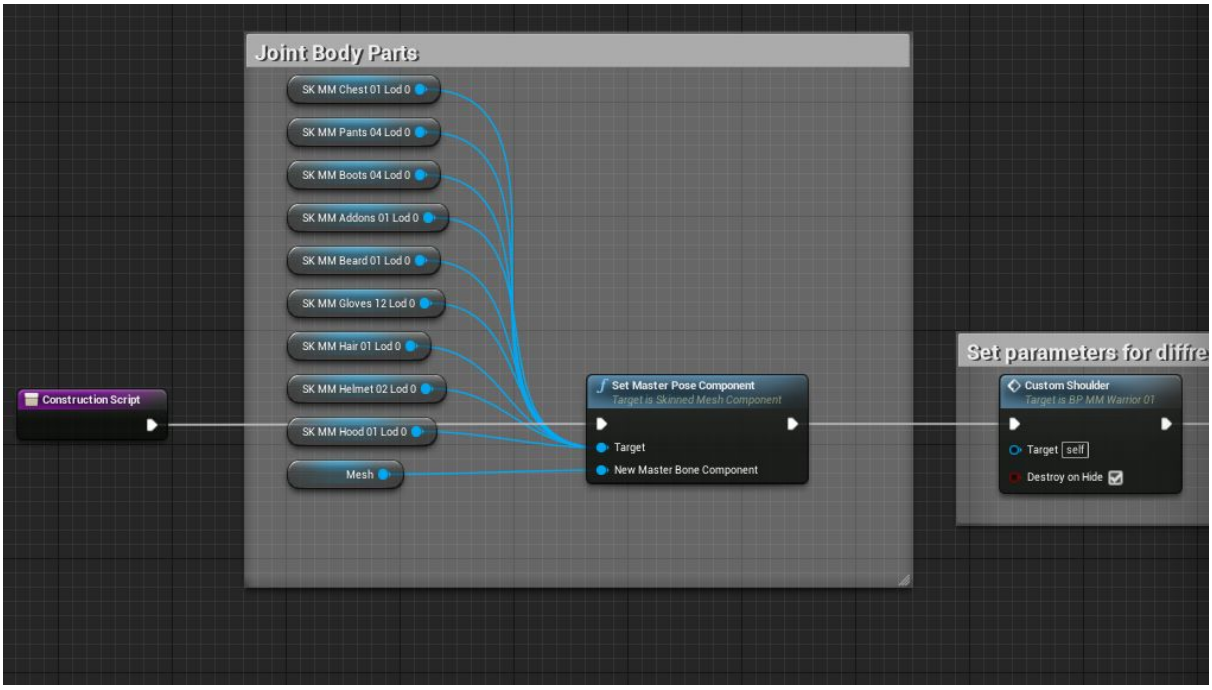
# Blueprints

## BP_MedievalCitizen_01

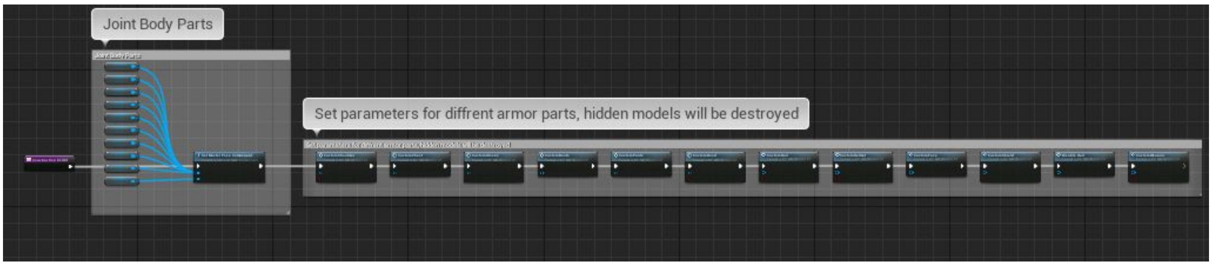
This BP inherit from Character class - you can use this for your player OR you can change parent class to your custom character class - then it will inherit from your class and by your class also from Character class.

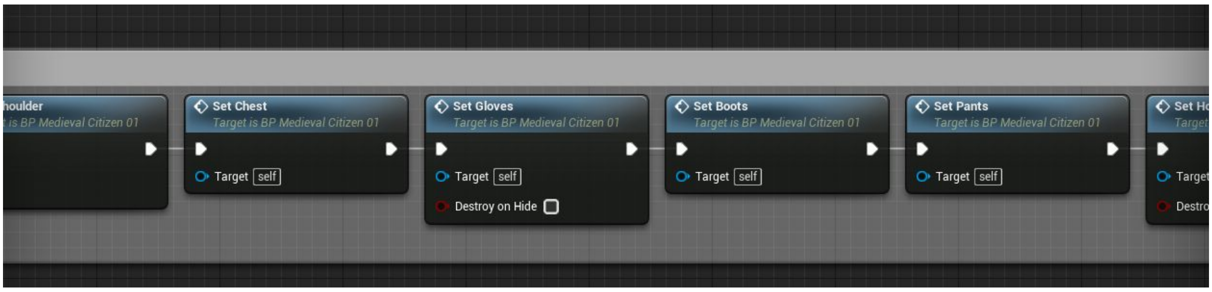
### Construction Graph

In the beginning of construction script, BP join different skeletal meshes to be controlled by one skeleton. (BP_PickUpItem is also using this node to join models placed in the world)

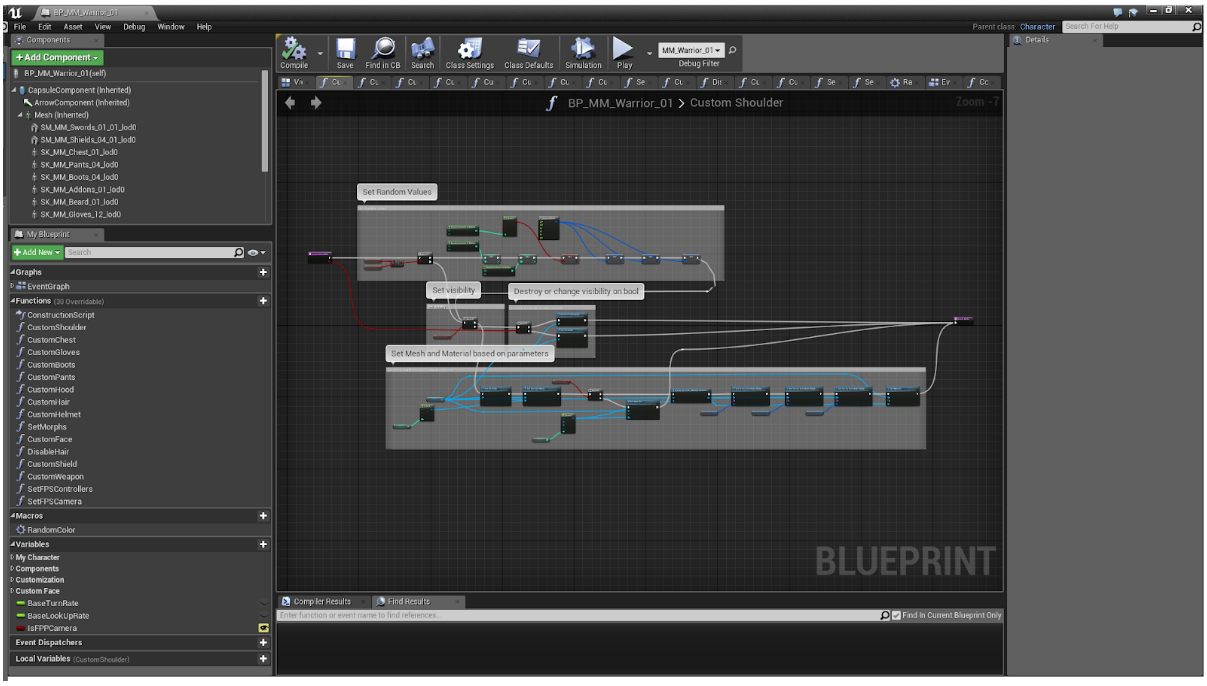



Each body parts have his own variables to control randomization. Construction script launch each function responsible for each body part.

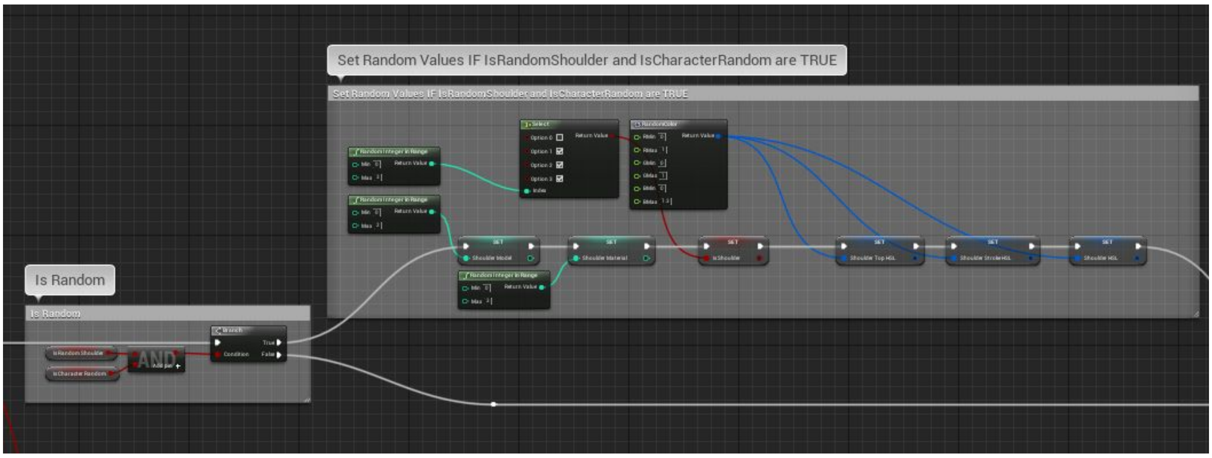




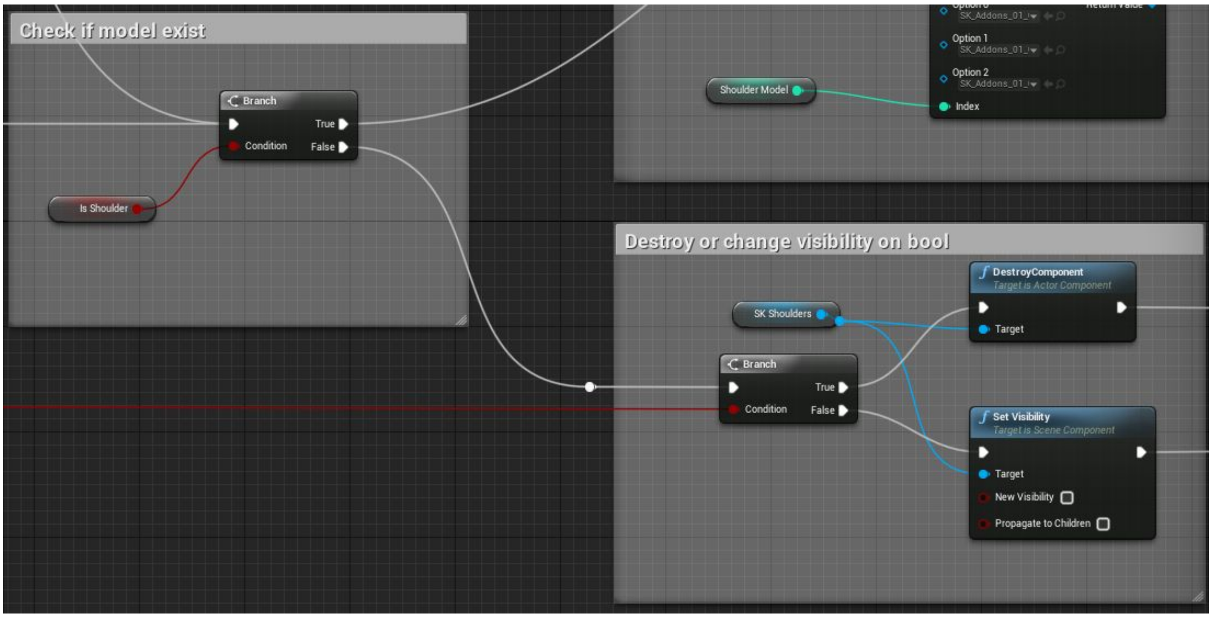

**Functions(SetShoulder, SetChest, SetGloves, SetBoots, SetPants, SetHood, SetHair, SetHelmet, SetShield, SetWeapon, SetApron)**
Are divided into two or three parts:



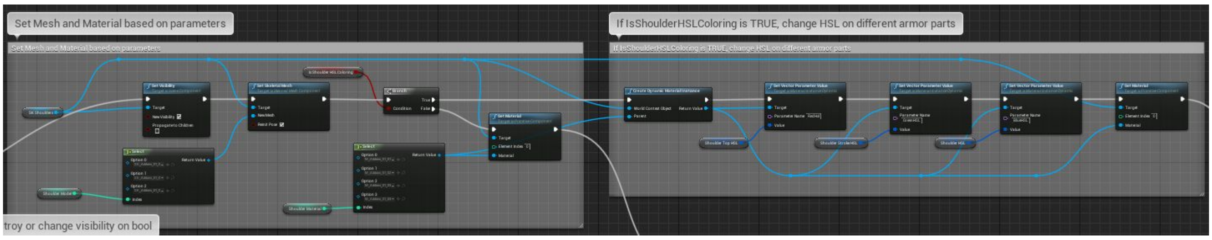

1. Generate random variables.



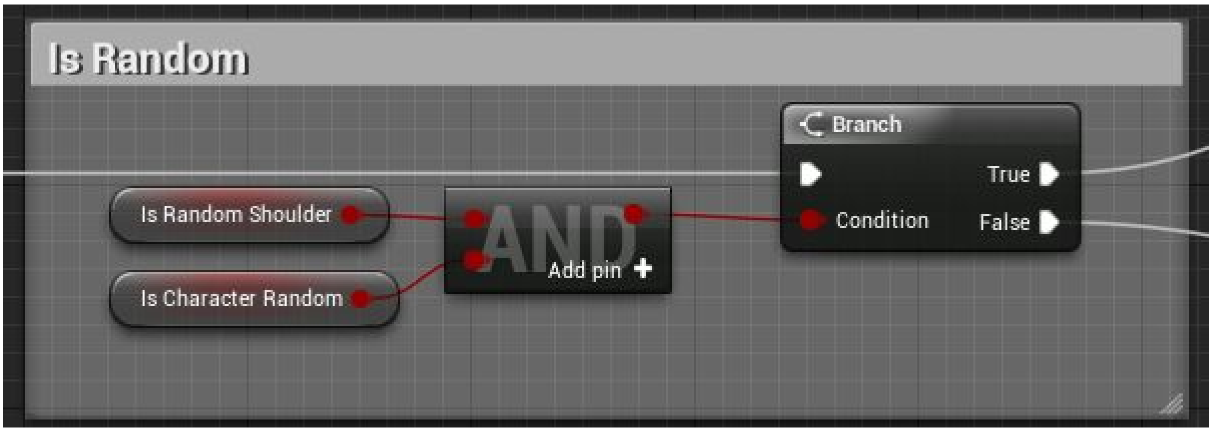

2. Set visibility (only shoulders, gloves, boots, hood, hair/beard, helmet, shield, weapon)




3. Set mesh and material instance parameter based on variables ( it doesn't have to be random, you can change them also manually)



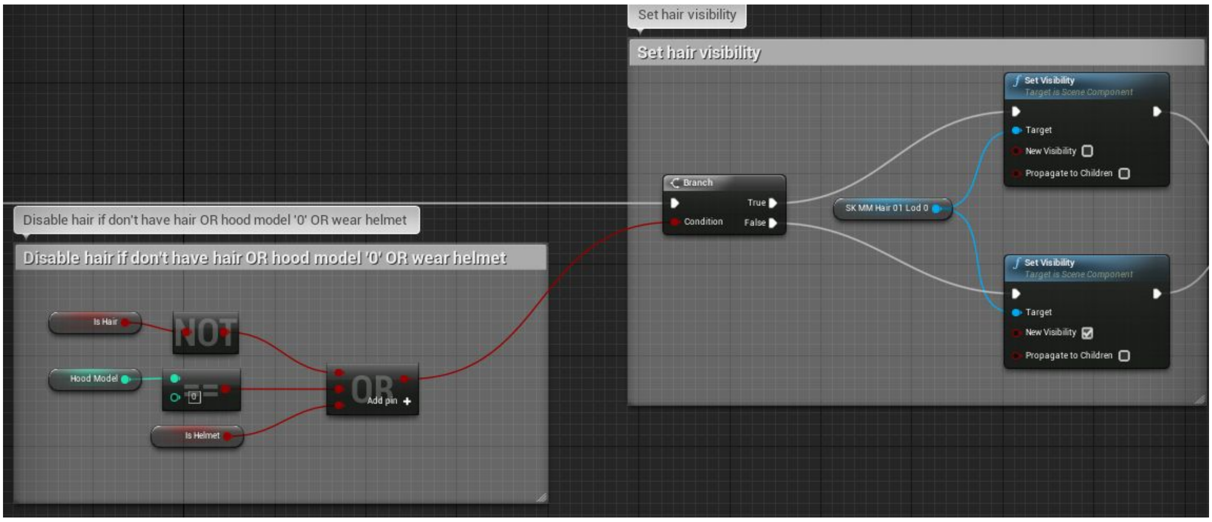

Variables random depends on 2 bools:




Is Random “Item” - e.g randomise everything except chest and shield (to create random character with one common characteristics)
Is Character Random - to control if the character will randomise variables in every recompile (note: any little change in character - for ex. moving character will recompile BP)

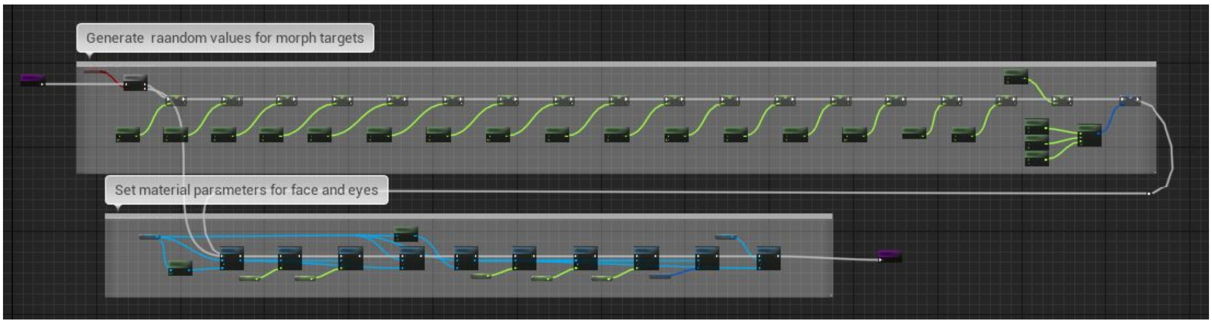

“Destroy on Hide” bool will control if the model should be destroyed. It’s better to destroy models preventing some of the calculations. But if you want to randomize character in Play model I suggest you to leave these on OFF (model will be hidden), OR make sure character is fully equipped before play. “Destroy on Hide” is not so important.

**DisableHair function:**

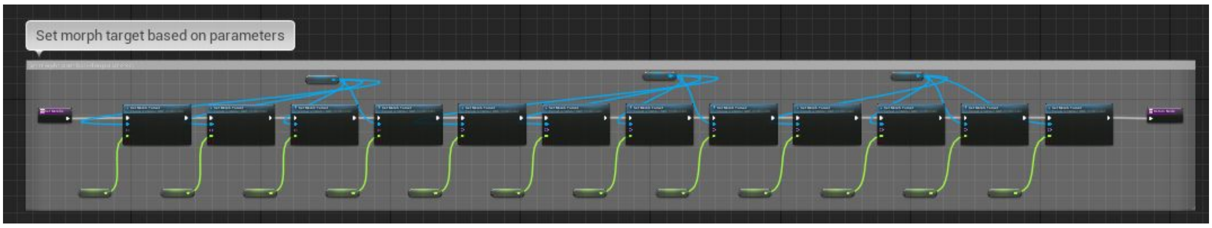



It's used for hide hair if character have helmet or specific hood model ( nr 0 ). You can use this node also after character pick helmet (or hood model nr 0)

**CustomFace function:**

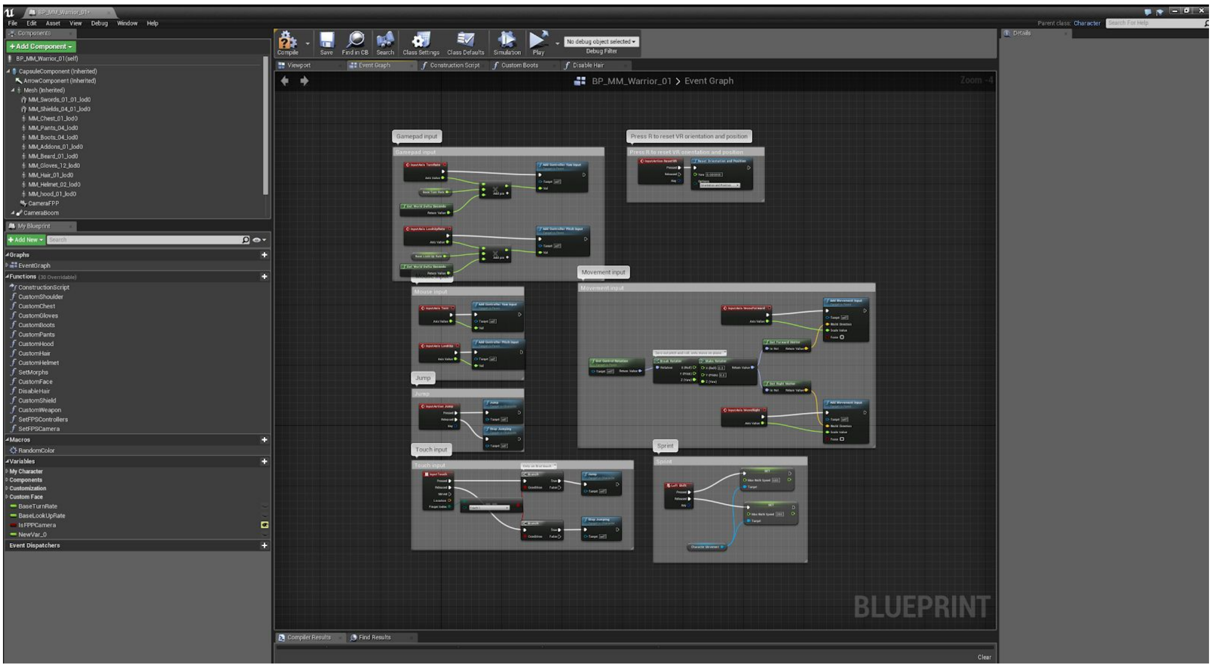



CustomFace generate random values for materials instance and morph target then set material instance parameters. Values for morph target are used in **SetMorphs function:**

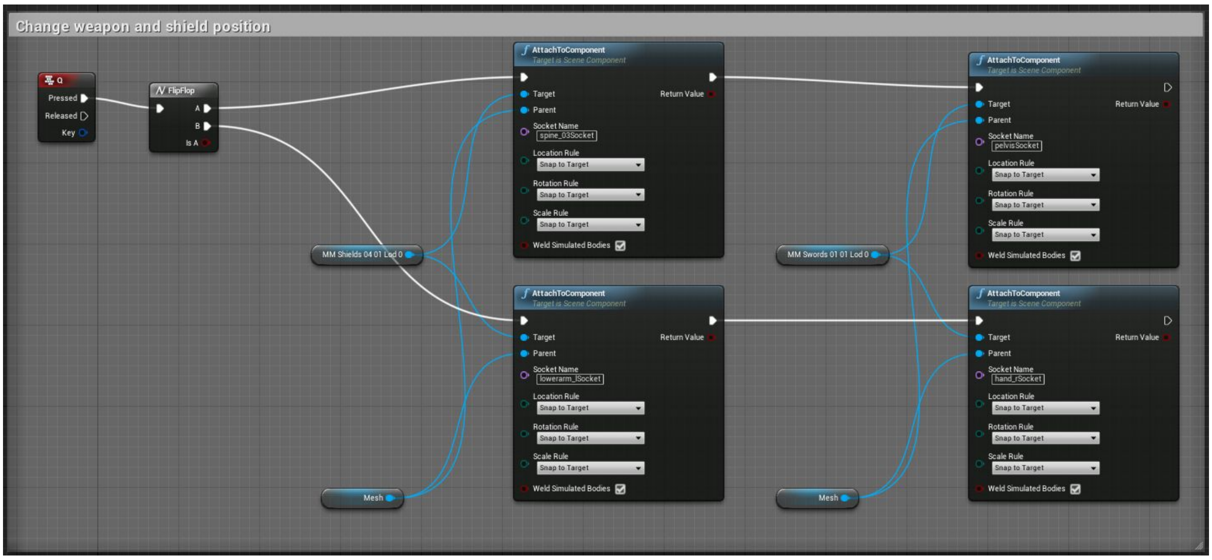


## Event Graph

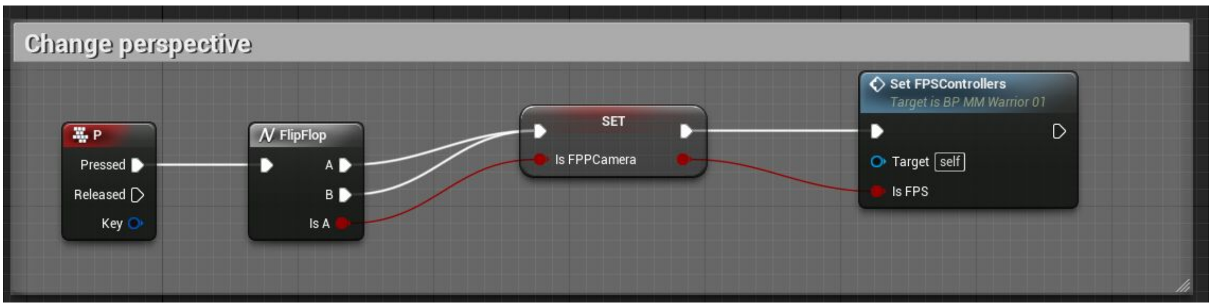
Standard Epic Third Person Character nodes:




"Q key" will change weapon and shield position to different socket.



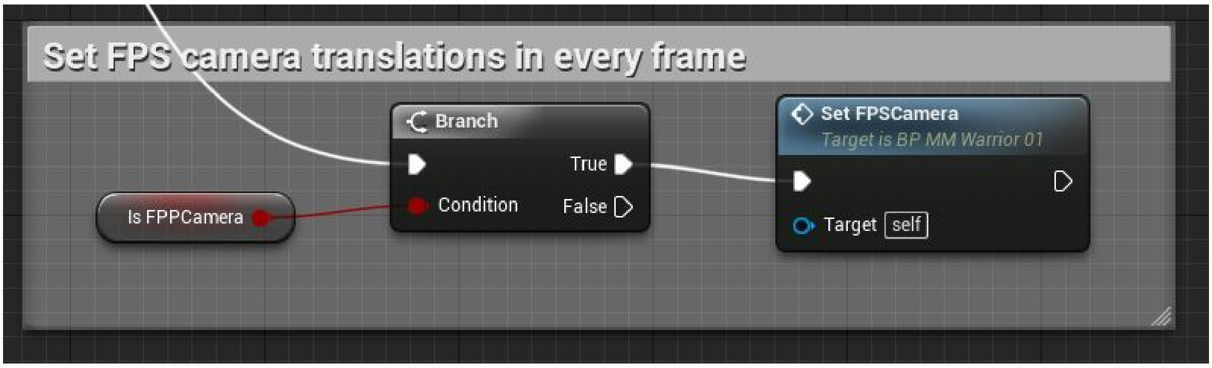

"P key" will change character perspective



Set FPS Controllers function change Character variables:

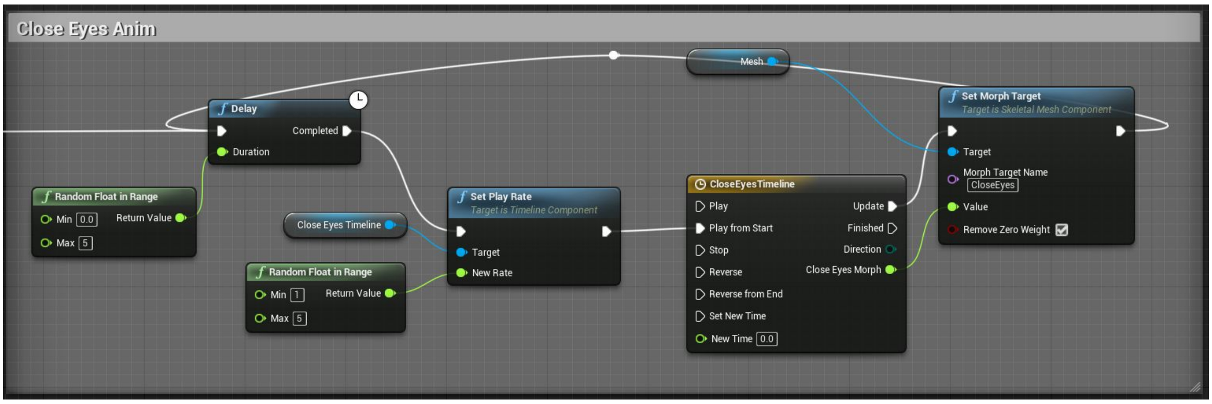
- Use Controller Desired Rotation
- Orient Rotation to Movement
- Set Active Camera
- Use Controller Rotation Yaw

It also set camera correct position in every Tick  (to have nice camera shakes - but not to much)

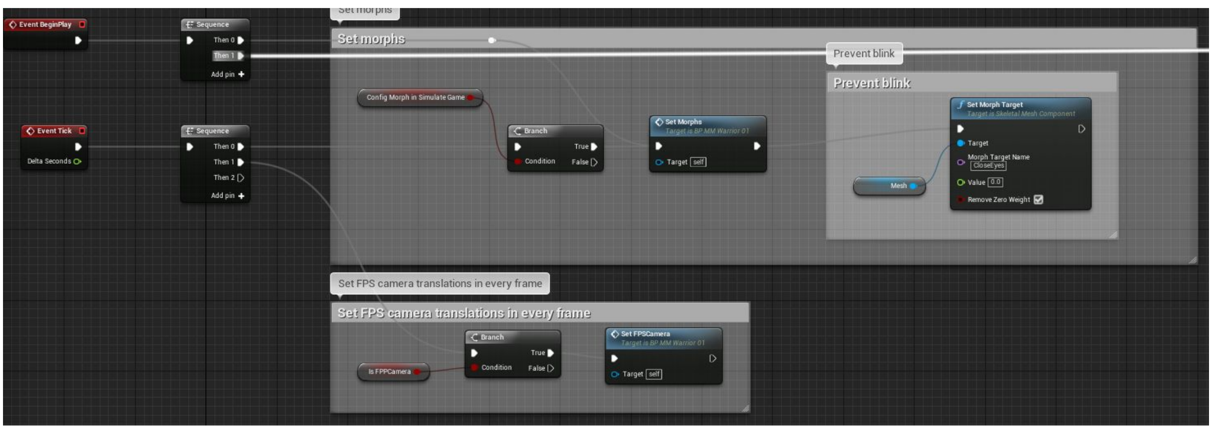



Close Eyes Animation:

Character in random delay with random play rate use morph target to blink character eyes.




Face morphs




There are few very important things to remember:

1. In UE4 morph target need Tick to update - it can’t be setup in construction graph :|

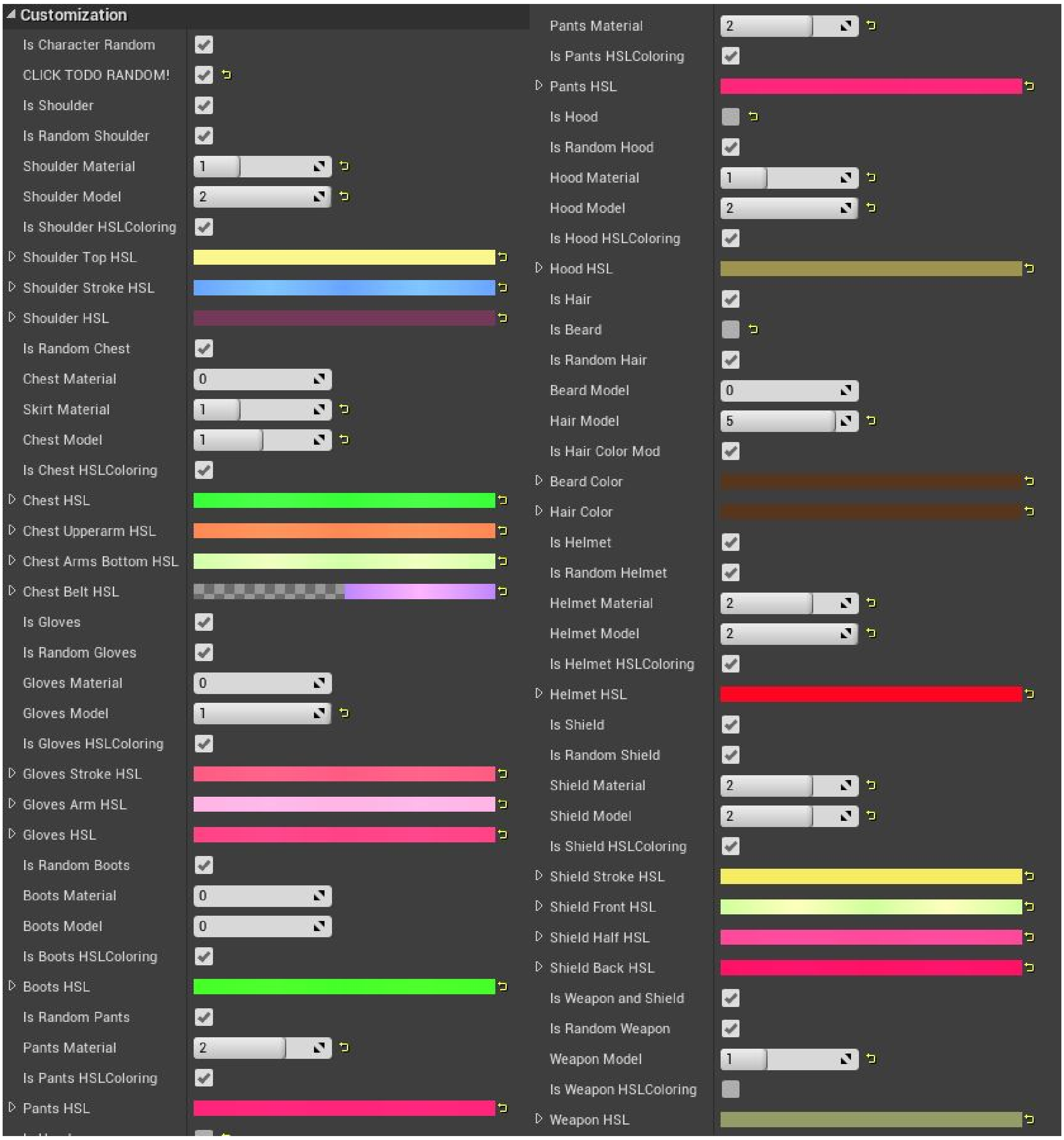
2. If you are changing face shape  in simulation you have to set Config Morph In Simulate Game to ON to command UE to refresh morphs.


3. If you are not changing face shape in runtime it's important to disable Config Morph In Simulate Game to OFF (for better optimization) .

Some of the screenshot are from Modular Medieval Warriors 01 - these blueprints works the same in Modular Medieval Citizens.

# Character creator

## Clothes/armor customization



Is Character Random - set this to OFF to stop randomise (on every recompile construction graph).

CLICK TODO RANDOM! - this bool does nothing :) but if you change any variable it always force BP to recompile, in this case it will generate new character.

Is “Item” - set if this item is visible (only for: shoulders, gloves, helmet, hood, hair, beard, shield, weapon)

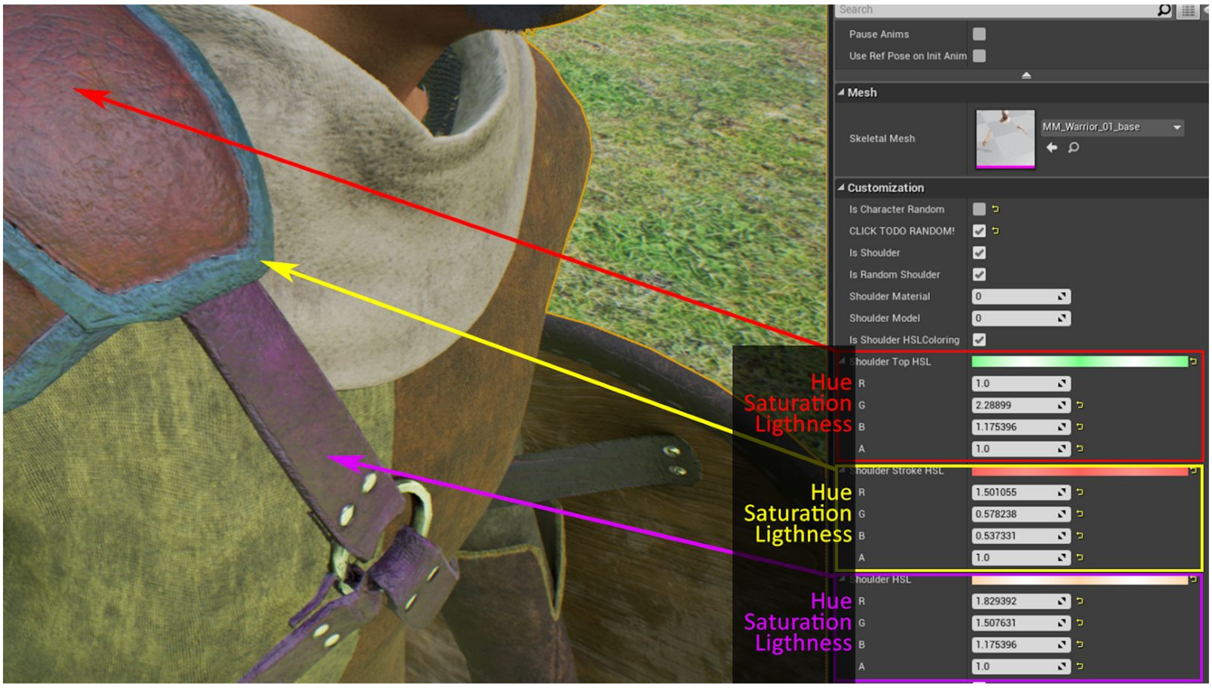
Is Random “Item” - if you set this to OFF, blueprint in recompile will NOT generate random variables for this type of item (you can create random character with similar parts OR change one type of item manually and leave rest to random).


“Item” Material - change material for this item

“Item” Model - change model for this item

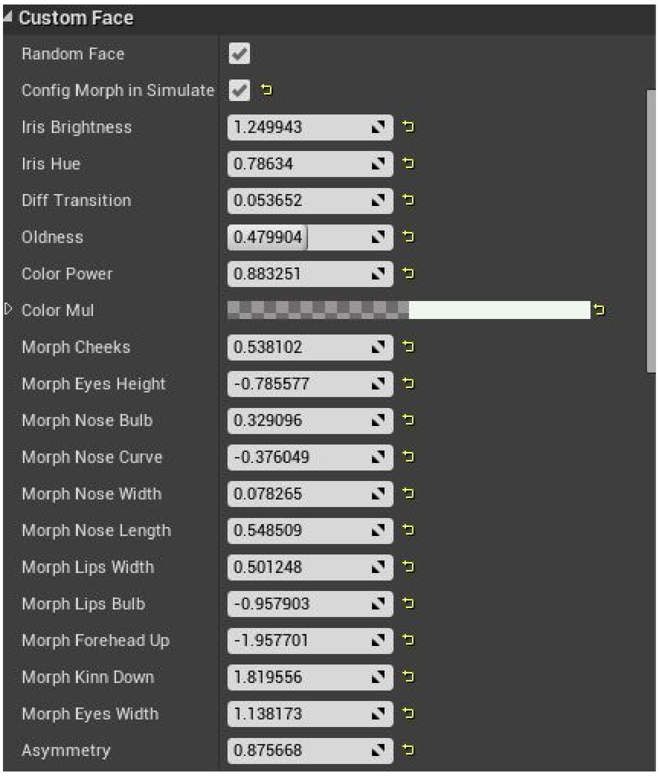
Is "Item" HSL Coloring - set this to OFF to disable item coloring using Hue, Saturation, Lightness transformation.


"Item" HSL - R - Hue, G - Saturation, B - Lightness




*I decided to use HSL instead of Color Multiply - HSL gives you more control and do not replace texture information.*

# Face creator



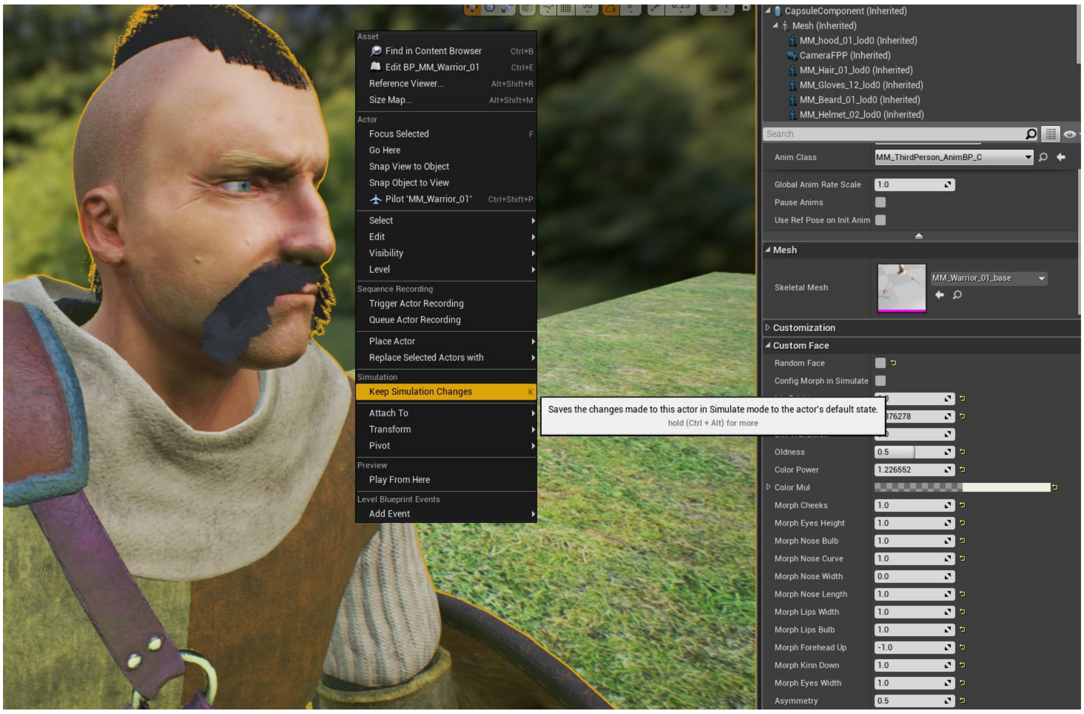

Random Face - set this to OFF to prevent BP from randomising variables

Config Morph in Simulate Game - set this to ON while you are changing morphs in simulation. Set this to OFF when you are happy with morph setting. (it will prevent UE from updating morph in every frame)

Iris Brightness - self explanatory
Iris Hue - self explanatory
Diff Transistion - controll transition between diffuse 1 and 2
Color Power - change color contrast using power
Color Mul - color multiply

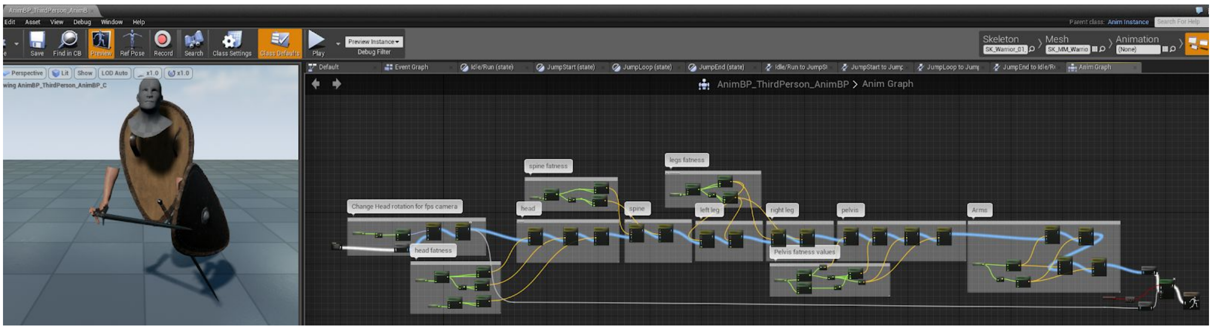
Morphs:
It's important to know that UE need tick to set morph targets. You can't set up morph's in construction graph.

The best way to change morph variables is to set them in simulation mode.
**Important: After changing variables press "K" or Right Mouse - choose "Keep Simulation Changes"**

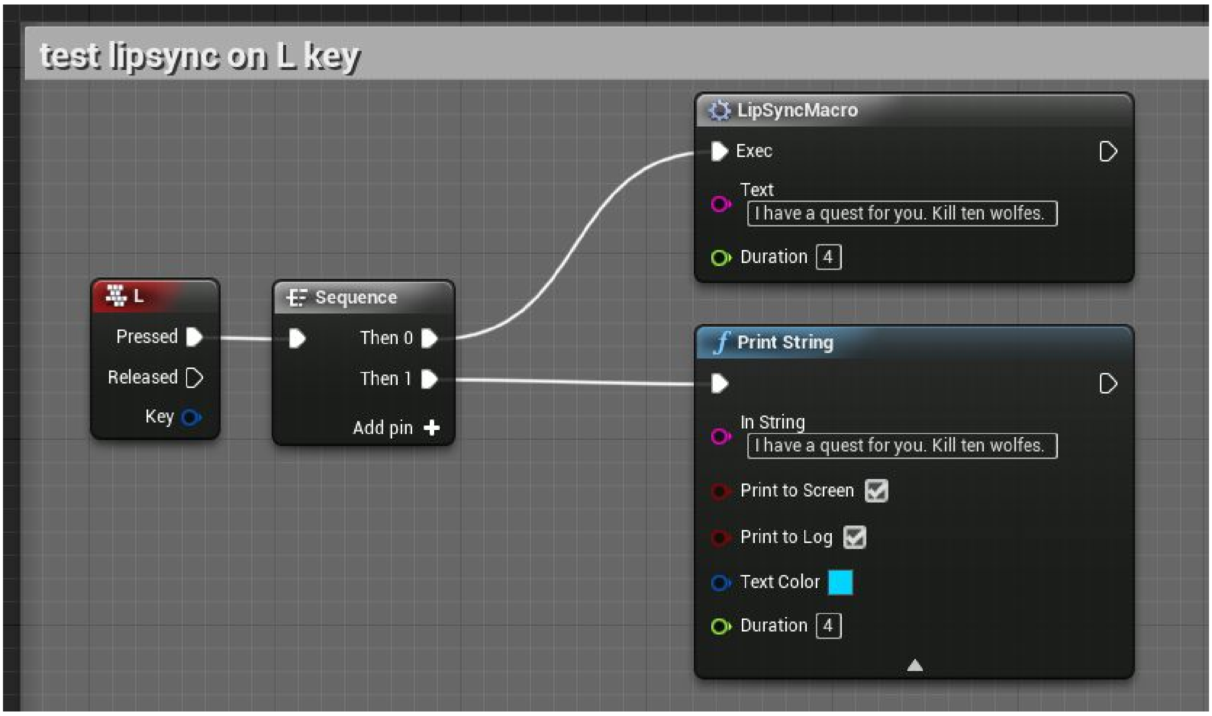



# Body Proportion -

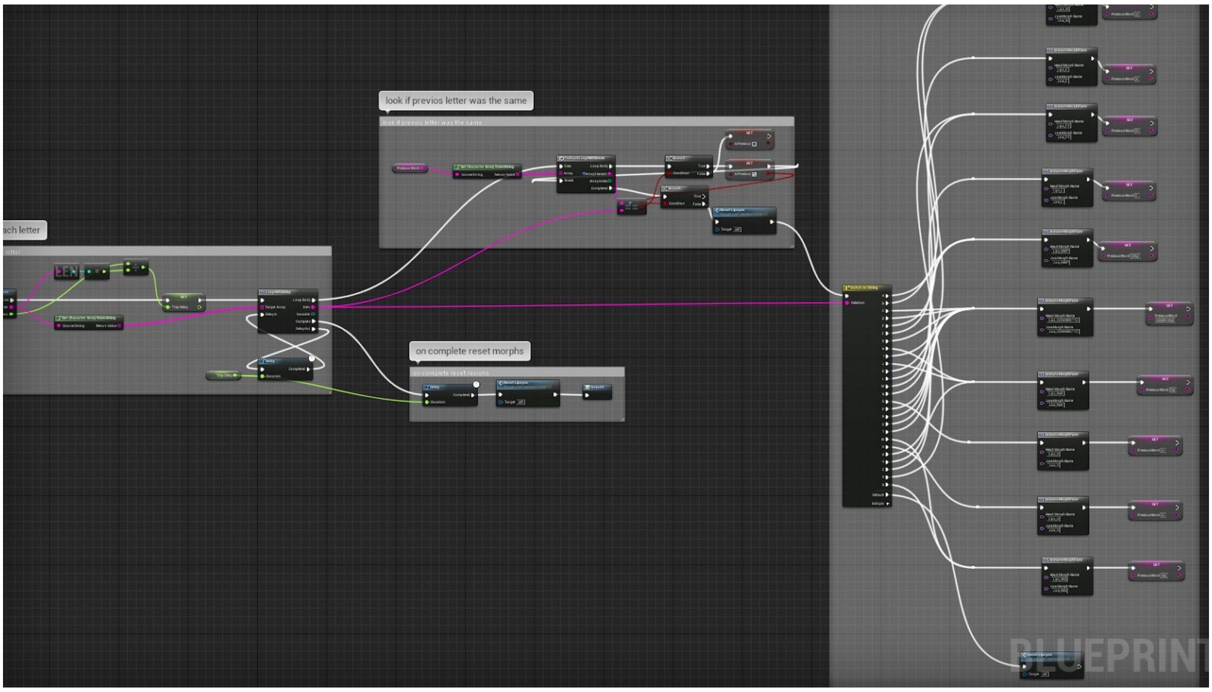

Body Build scales spine character, Scale Arms scales arms. To scale skeleton I used Animation BP. If you have your own animation BP, you can paste nodes from Anim Graph:



# LipSyncMacro - simple lip sync system:




This marcro, for each letter will change face morph target. 10 blendshapes:
AI, E, CDGKNRSTYZ, FV, O, U, L, MBP,  WQ, rest
Input: String and Duration.

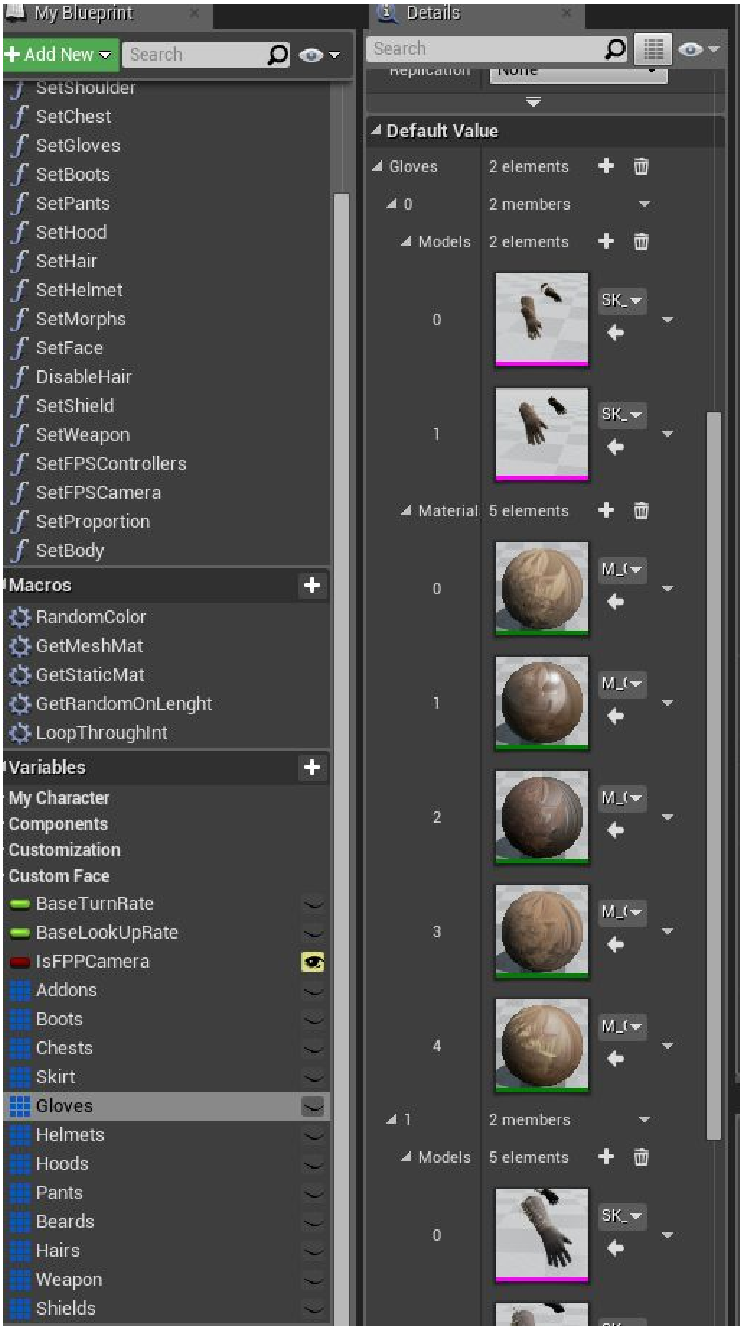



This lip sync system is for simple projects - it is not the best. There is 10 blendshapes you can use for your own system.

# Add custom mesh to BP

If you want to add your own meshes and materials to customization script.

1. Open BP and find appropriate structure: like gloves
2. Add child in structure root
3. Add models in array
4. Add materials in array

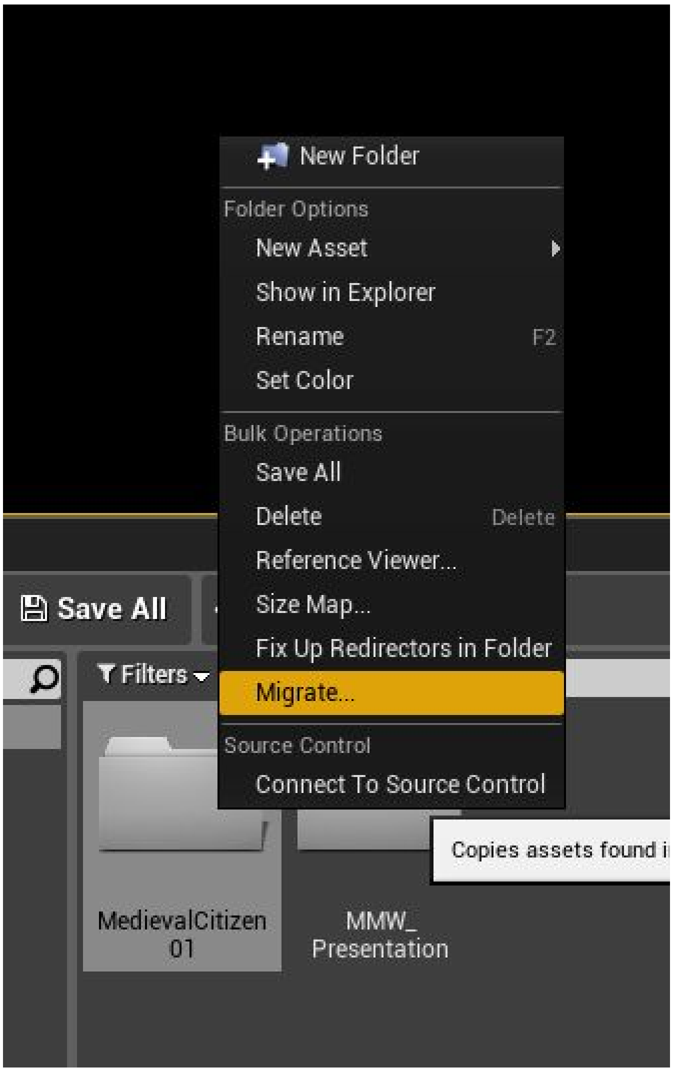


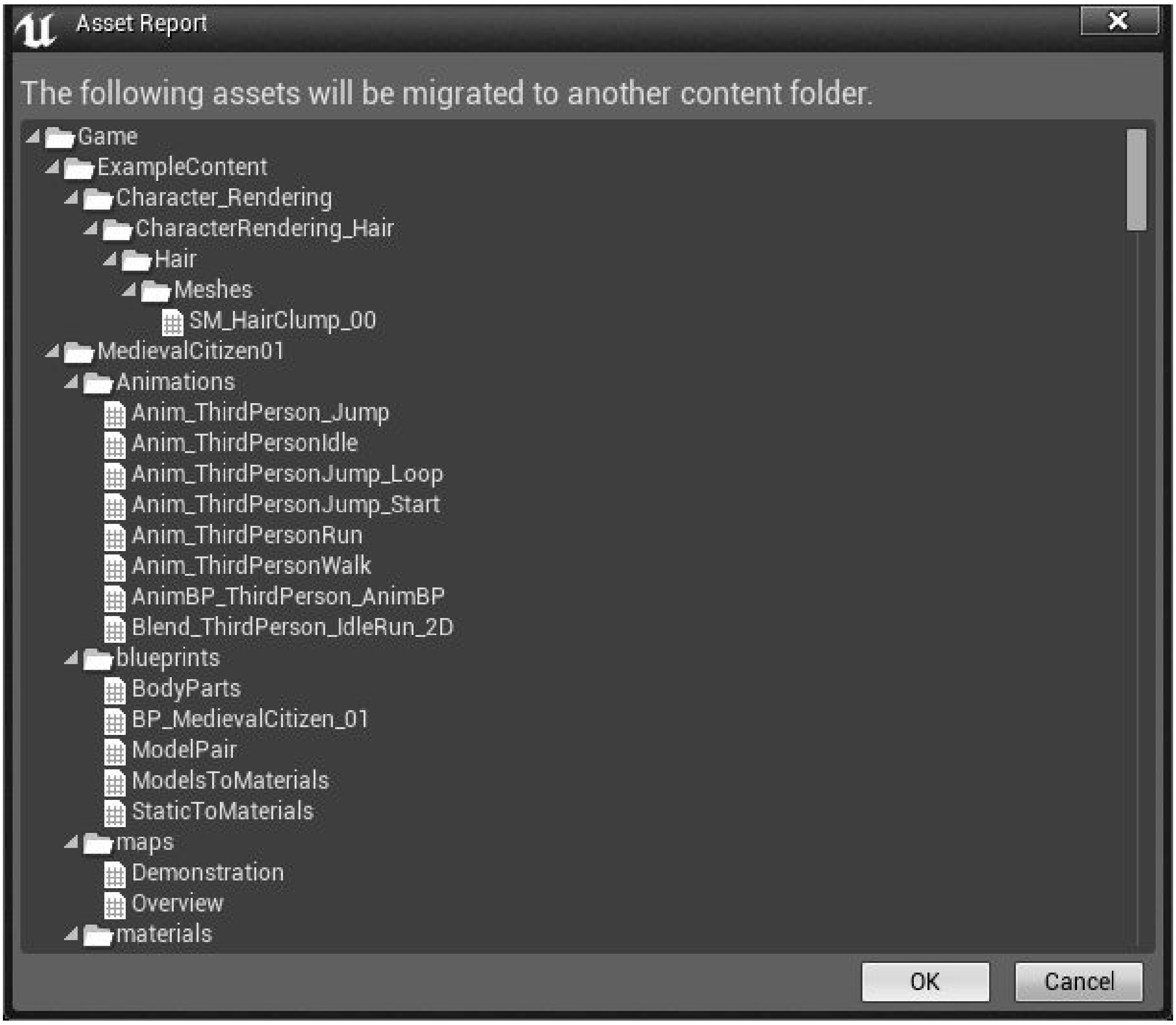
# Merging packages

With documentation file you will find additional blueprint that will work after importing Modular Medieval Warrior and Modular Medieval Citizen as one BP: BP_MedievalCharacters_01.

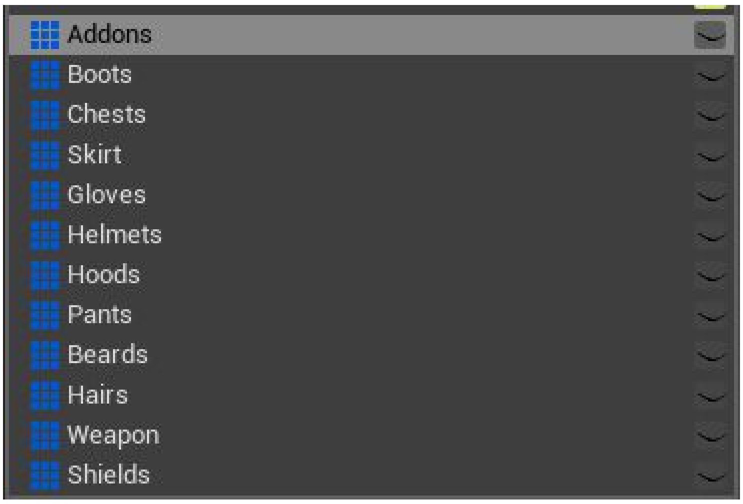
You can also make it work manualy:

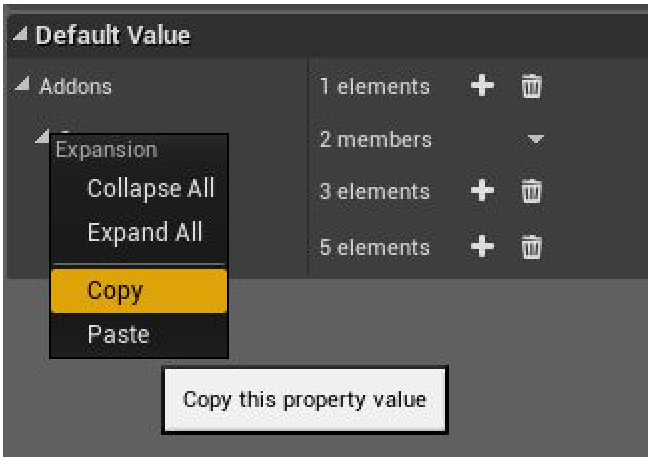
1. Create NEW third person character project or use your OWN.
2. Open Modular Medieval Citizen.
3. Migrate main folder to NEW project Content folder



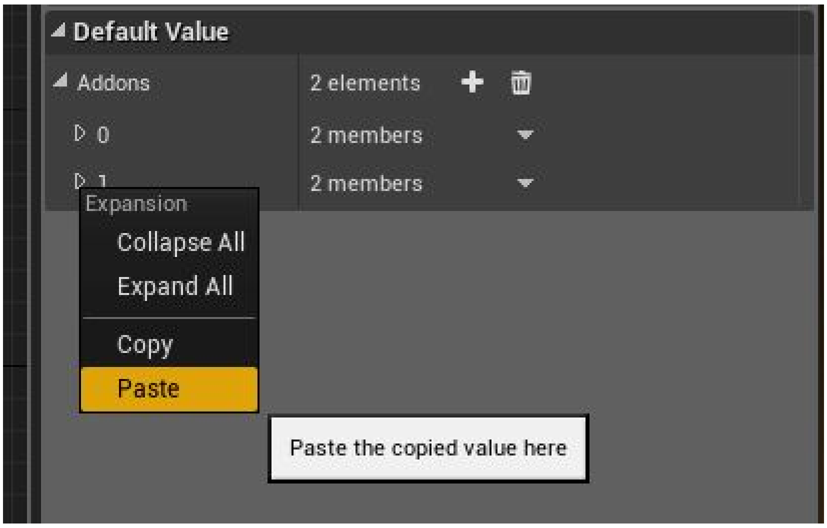



4. Click ok and select Content folder from your project.
5. You can also add to your project using Epic Library. (skip steps 2-4)
6. Do steps 2 ,3 ,4 or 5 for every other Modular Medieval projects.
7. Duplicate BP_MedievalCitizen_01 and name it BP_MedievalCharacter
8. Open main BPs per every project like:  BP_MedievalCitizen_01, BP_MedievalWarrior_01,.
9. Copy array elements from BP_MedievalWarrior_01.






10. Add array element in BP_MedievalCharacter and paste this element.

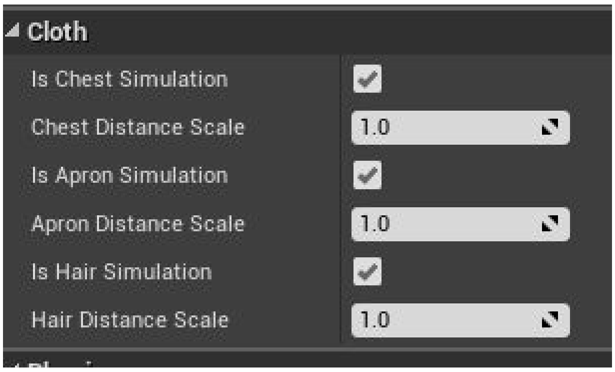



11. Do this for every array (addons, boots, chests.. etc.)

# Issues

## FPS

If you fell fps dropping on Lod0 it because of cloth simulation. You can easily disable cloth using these parameters.




Also check if you have turn OFF Config Morph in Simulation Mode - If it’s ON it will take a lot of FPS, use this only during creating character in simulation mode.,

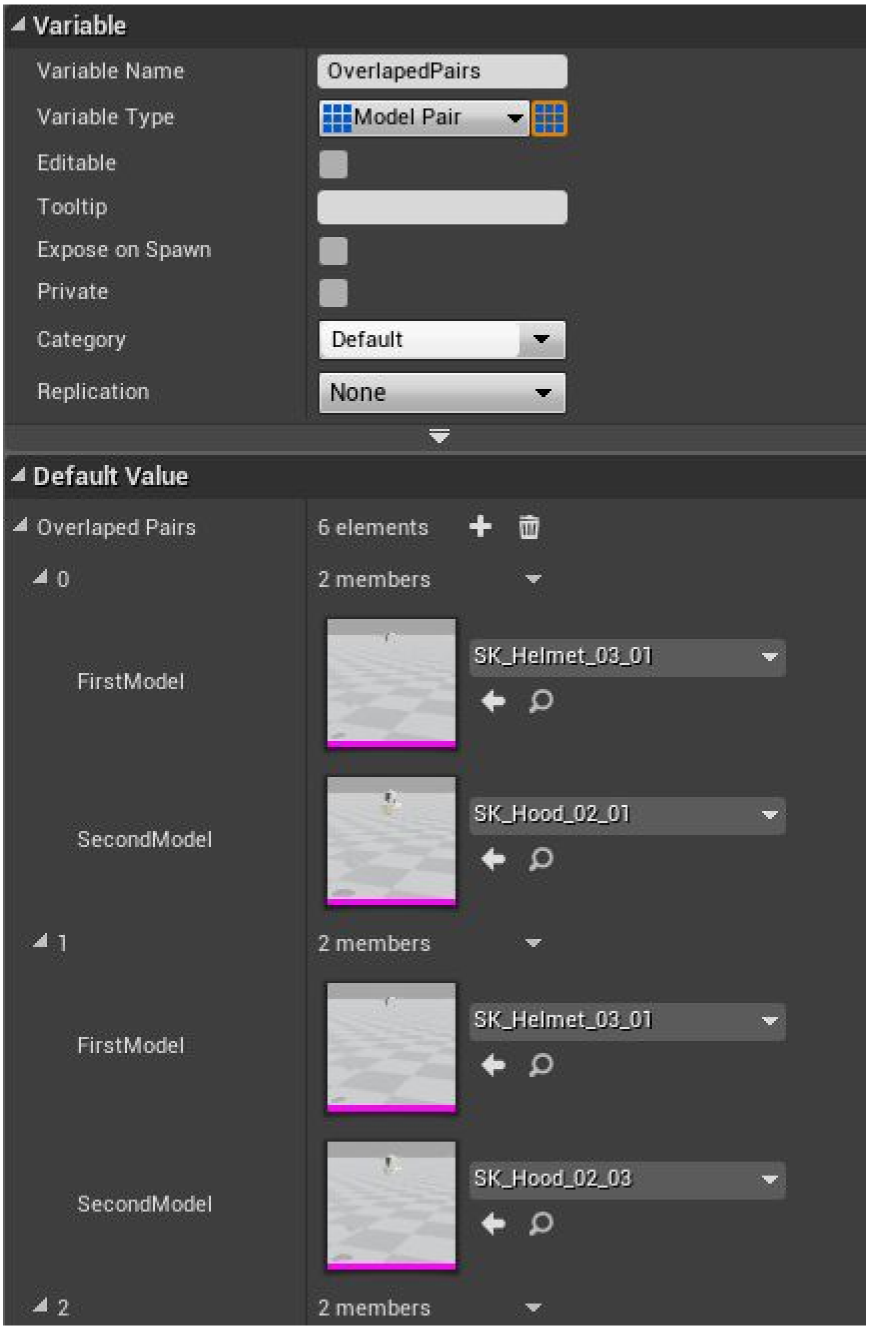
## Cloth

Hair nr5 may look a little bit crazy, but after simulate it will look better.


This hair may act weird - but I personally like it, so I added this to the package. (this is only hair type that need cloth to look right)

## Overlapped meshes

If you have overlapped meshes after migrating Warrior package or after you add your own mode,. you can add pairs that interfere with each other to stop the generation of such pairs in randomization script using OverlapedPairs array




## Lips

If you play game in Editor mode without compiling BP_MedievalCitizen_01, lips will act weird - In standalone/ packaged mode or after compile BP, it will work correct.

# Contact

If you have any question, feel free to contact me at Assets3d <3dasset@gmail.com>
Facebook: https://www.facebook.com/assets3d/
Youtube: https://www.youtube.com/channel/UCP-Y5dj9apJBepq2U_R6JwQ
Homepage: http://assets3d.com/

# License

Modular Medieval Citizens 01 pack can be purchased via Unreal Engine Marketplace. All 3d models can be used for commercial purposes only by owners who bought this package. The sharing or selling of package data is strictly prohibited.